RGPV PREVIOUS YEAR QUESTIONS (PYQ) ANALYSIS REPORT

Course: B.Tech. I/II Semester • Subject: Engineering Physics (BT-201 GS)

Comprehensive Unit-Wise Analysis & Exam Guide

Analyzing Exam Sessions: June 2022 – December 2025

Prepared For: B.Tech. First-Year Students (Grading System)

University: Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Madhya Pradesh

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Section 1: Unit-wise PYQs

This section provides a complete, word-for-word extraction of all questions from the RGPV Engineering Physics (BT-201 GS) papers from 2022 to 2025, categorized systematically into the five primary curriculum units.

Unit 1: Quantum Mechanics & Wave Nature of Matter

1. Explain the wave nature of particles. Derive the time-independent Schrödinger equation for a particle in one-dimensional potential box and write its wavefunction. **(Dec 2025)**
2. A particle of mass $9.11 \times 10^{-31}\ kg$ is moving in a one-dimensional box of width 10 nm. Calculate the energy of the particle in the first three excited states. **(Dec 2025)**
3. Discuss the uncertainty principle. Derive the relation between group velocity and phase velocity for a free particle. **(Dec 2025)**
4. Derive time dependent and time independent Schrodinger wave equation. **(June 2025, June 2023, June 2022)**
5. Discuss the physical significance of wave function. **(June 2025, June 2022, Nov 2022 [Short Note])**
6. Deduce the energy eigenvalues and wave function of a particle moving in one dimensional box. **(Dec 2024, June 2022)**
7. Explain about the Heisenberg's uncertainty principle in detail. / State and Prove. **(Dec 2024, June 2024, Dec 2023, June 2023 [Elementary proof], Nov 2022, June 2022 [Short Note])**
8. Discuss about the Free Particle Wave Function and Wave Packets. **(Dec 2024)**
9. Define phase velocity and group velocity. **(June 2024)**
10. Derive the time independent Schrödinger wave equation. **(June 2024)**
11. Discuss the energy and momentum operator. / Derive. **(June 2024, Dec 2023, June 2023)**
12. Explain the Dual nature of matter / light. **(June 2024, Dec 2023)**
13. Prove that for one dimensional motion of a particle in a box $\psi_n = A\sin(n\pi x/L)$. **(June 2024, Dec 2023)**
14. Deduce the relation between phase and group velocities. **(Nov 2022)**

Unit 2: Wave Optics (Interference & Diffraction)

1. Describe how Newton's ring experiment can be used to determine the refractive index of a liquid. **(Dec 2025)**

2. Explain briefly the Rayleigh criterion of resolution. Discuss the resolving power of plane transmission grating and find the relation between resolving and dispersive power of the grating. **(Dec 2025)**
3. Discuss the phenomenon of Fraunhofer diffraction at a single slit. Show that the intensity of first subsidiary maximum is about 4.5% of the principal maximum. **(Dec 2025)**
4. Newton's rings are observed normally incident in reflected light of wavelength 6000 Å. The diameter of the 10^{th} dark ring is 0.50 cm, find the radius of curvature of lens and thickness of the film. **(Dec 2025)**
5. Explain Michelson interferometer and derive expression for fringe shift due to displacement of mirror. / Explain working. **(Dec 2025, June 2025, June 2024, Dec 2023, June 2022)**
6. Describe the construction and working of Mach-Zehnder interferometer. **(June 2025, June 2023, Nov 2022, June 2022 [Short Note])**
7. What is superposition of wave and interference of light by amplitude splitting? **(June 2025)**
8. What is the importance of Rayleigh criteria. Explain Rayleigh criterion for limit of resolution. **(June 2025, Dec 2023)**
9. Explain the Fraunhofer diffraction due to single slit with necessary analysis. / Derive expression for intensity. **(Dec 2024, June 2024)**
10. How the Newton's rings are formed? Deduce the expression for diameter of dark and bright fringes. **(Dec 2024, Nov 2022)**
11. Interference and Young's double slit experiment / Explain fringe width. **(Dec 2024 [Short Note], Dec 2023, June 2023, Nov 2022 [Short Note])**
12. Define path difference and phase difference. **(June 2024)**
13. What should be the minimum number of lines in a grating which will just resolve in the second order, the lines, whose wavelengths are 5890Å and 5896Å. **(June 2024)**
14. Differentiate between division of amplitude and division of wavefront. **(June 2024, Dec 2023)**
15. In Newton's ring experiment the diameter of nth and $(n+14)^{th}$ rings are 0.42 cm and 0.70 cm respectively. If the radius of curvature of plano convex lens in 100 cm. Calculate the wavelength of light. **(June 2024, Dec 2023)**
16. Newton's ring. **(June 2023 [Short Note])**
17. Explain the formation of Newton's rings. Obtain the expression to find the wavelength of light in Newton's rings experiment. **(June 2022)**
18. Explain about the diffraction grating. A parallel beam of Sodium light incident on plane transmission grating having 4250 lines per centimeter and a second order spectral line is observed at an angle of 30°. Find the wavelength of Sodium light. **(Nov 2022)**
19. Define superposition of waves. **(Dec 2023)**

Unit 3: Semiconductor Physics & Solid State

1. Explain Hall effect and derive expression for Hall voltage. Discuss its application. / Hall coefficient and Hall angle. **(Dec 2025, June 2025 [Short Note], Dec 2024, June 2023, June 2022, Nov 2022)**
2. Explain Fermi level for intrinsic and extrinsic semiconductors. Discuss the density of states and Bloch's theorem. **(Dec 2025)**
3. Explain the variation of Fermi level in N-type semiconductor with concentration and temperature. / Shifting. **(Dec 2024, June 2024, Dec 2023, June 2022)**
4. What is P-N junction diode, explain its working and discuss its V-I Characteristics. **(Dec 2024, June 2023, Nov 2022)**
5. Write short notes on "Density of State". **(Dec 2023)**
6. State and prove Bloch Theorem. **(Dec 2023, Nov 2022 [Short Note])**
7. What is Zener diode? Discuss its V-I Characteristics. **(June 2025, June 2024, June 2023, June 2022)**
8. Discuss the free electron theory of metals. **(June 2024)**
9. Explain Kronig Penney model for periodic potential. Write down the Schrodinger equation and discuss conclusion. **(Dec 2024 [Short Note], June 2023)**
10. Solar cell. **(June 2025 [Short Note], June 2023 [Short Note], June 2022 [Short Note], Nov 2022)**
11. Intrinsic and Extrinsic semiconductor through Fermi level. **(June 2023 [Short Note])**
12. Differentiate between Avalanche and Zener breakdown. **(June 2023)**

Unit 4: Lasers & Fiber Optics

1. Derive the relation between Einstein's coefficients (A & B) and explain the concept of population inversion. **(Dec 2025, June 2025, Nov 2022)**
2. Explain the construction and working of He-Ne laser with suitable diagram. **(Dec 2025, Dec 2024, June 2024, Dec 2023, Nov 2022)**
3. Calculate the NA, Acceptance angle, Critical angle and acceptance cone, for a fiber have RI of 1.55 and 1.50 respectively for its core & cladding. **(Dec 2025)**
4. Explain acceptance angle and acceptance cone of optical fiber. Derive expression for them. / Explain Numerical aperture. **(Dec 2025, June 2025, Dec 2024, June 2024, Dec 2023, June 2022)**
5. Explain the construction and working of Ruby laser with neat diagram. **(June 2025, June 2024, June 2022)**
6. Explain Construction and working of CO_2 LASER with suitable energy level diagram. **(June 2025, June 2023)**

7. What is LASER? Discuss the Properties of LASER light. **(Dec 2024, June 2024, Dec 2023, June 2022, Nov 2022 [Short Note])**
8. Find the intensity of laser beam of 10 mW power and having diameter of 1.3 mm, assuming the intensity to be uniform across the beam. **(June 2024)**
9. Write down the applications of LASER in Engineering and Medicine. **(June 2024, Dec 2023, June 2023)**
10. Write down the importance of total internal reflection in optical fiber. **(June 2024, Dec 2023)**
11. Calculate the numerical aperture and acceptance angle for an optical fiber, given that the refractive indices of core and cladding are 1.45 and 1.41 respectively. **(June 2024, Dec 2023)**
12. Calculate the numerical aperture of a fiber with refractive indices of core and cladding are 1.55 and 1.50, respectively. **(Nov 2022)**
13. Find the acceptance angle of a fiber having core and cladding refractive indices are 1.75 and 1.70 respectively. **(Dec 2024)**
14. Find the acceptance angle of a fiber having core and cladding refractive indices are 1.55 and 1.50, respectively. **(June 2022)**
15. What is the working principle of an optical fibre? How light propagates through it? Define terms: Numerical aperture, Population inversion, V-number. **(June 2023)**
16. V-number. **(Dec 2024 [Short Note], June 2022 [Short Note])**
17. Write down the difference between spontaneous and stimulated emission. **(June 2023)**

Unit 5: Electrodynamics & Vector Calculus

1. Prove the Poynting theorem in electrodynamics and explain the physical significance of each of the term appearing in the final expression of the theorem. **(Dec 2025)**
2. Poynting vector. **(Dec 2024 [Short Note])**
3. Derive Maxwell's equations in vacuum / non-conducting medium. **(Dec 2025, June 2025, June 2024, Nov 2022)**
4. Starting from the principle of charge conservation, derive the continuity equation for current densities in a conducting medium. **(June 2025, Dec 2024)**
5. Discuss about the continuity equation. / State and prove. **(June 2024, June 2023, June 2022)**
6. Write a note on different types of polarization in dielectric materials. **(June 2025, June 2022)**
7. Derive the Electric field and Electrostatic Potential for a charge Distribution. **(Dec 2024, Dec 2023)**
8. State and Derive the Divergence of Stoke's theorem and Gauss theorem. **(Dec 2024, Dec 2023, June 2023, Nov 2022)**
9. Define gradient of a scalar field and divergence of a vector field. **(June 2024, Dec 2023)**

10. Define the electric field intensity. Find out the expression for electric intensity due to infinite line charge. **(June 2023)**

Section 2: Important Topics

This matrix evaluates curriculum topics based on frequency trends from the 2022-2025 exam datasets, prioritizing high-yield areas.

Unit 1: Quantum Mechanics

Topic	Priority	Exam Analysis & Importance
Heisenberg Uncertainty Principle	★★★ High	Appeared in 6 separate sessions. Core foundational topic; mandatory to prepare statement, mathematical limits, and elementary proof.
Schrödinger Equation (1D Box)	★★★ High	Appeared 5 times. High derivation focus ($\psi = A \sin(n\pi x/L)$) often tied with numericals on energy states.
Phase & Group Velocity Relation	★★ Medium	Appeared 3 times. Standard algebraic derivation showing wavepacket behaviors.
Wavefunction Physical Significance	★★ Medium	Appeared 3 times. Focuses on Born interpretation and normalization constraints.

Unit 2: Wave Optics

Topic	Priority	Exam Analysis & Importance
Newton's Rings Experiment	★★★ High	Appeared 6 times. Regularly tests ring diameters, liquid refractive index, and numerical problem solving.
Michelson & Mach-Zehnder Interferometers	★★★ High	Appeared 7 times combined. Heavy focus on ray diagrams, fringe shift formulas, and applications.
Fraunhofer Diffraction (Single Slit)	★★★ High	Appeared 3 times. Demands complete intensity distribution formulas and proof of the 4.5% subsidiary maximum.
Rayleigh Criterion & Resolving Power	★★ Medium	Appeared 3 times. Covers transmission grating limits and resolving vs dispersive power links.

Unit 3: Semiconductor Physics

Topic	Priority	Exam Analysis & Importance
Hall Effect	★★★ High	Appeared 6 times. Essential derivation for Hall voltage and applications in carrier type determination.
Zener & P-N Junction Characteristics	★★★ High	Appeared 5 times. Standard V-I characteristics, breakthrough mechanisms (Zener vs Avalanche).
Fermi Level Shifting in N-Type	★★ Medium	Appeared 4 times. Focus on temperature and concentration variance equations.
Bloch Theorem & Kronig-Penney	★★ Medium	Appeared 4 times. Periodic potential insights; conclusions are frequently tested.

Unit 4: Lasers & Fiber Optics

Topic	Priority	Exam Analysis & Importance
He-Ne Laser Construction & Working	★★★ High	Appeared 5 times. High priority gas laser system with four-level energy transitions. Diagrams are mandatory.
Optical Fiber Parameters (NA & Acceptance Angle)	★★★ High	Appeared 6 times. Mathematical derivations and numerical calculations are universally asked.
Einstein Coefficients (A & B Relationship)	★★★ High	Appeared 3 times. Core thermodynamic equilibrium derivation for light amplification.
Ruby & CO2 Lasers	★★ Medium	Appeared 5 times combined. Solid-state vs molecular gas laser mechanism alternatives.

Unit 5: Electrodynamics & Vector Calculus

Topic	Priority	Exam Analysis & Importance
Maxwell's Equations in Vacuum	★★★ High	Appeared 4 times. Core set of 4 equations derived via differential/integral forms.
Equation of Continuity	★★★ High	Appeared 5 times. Conservation of charge framework; intermediate steps must be clean.
Vector Theorems (Gauss & Stokes)	★★ Medium	Appeared 4 times. Mathematical transitions between line, surface, and volume integrals.
Poynting Theorem & Significance	★★ Medium	Appeared 2 times. Explains electromagnetic power flux flow.

Section 3: Most Repeated PYQs

Identical and closely related recurring questions ranked by their precise appearance frequencies.

Unit 1: Quantum Mechanics

1. Heisenberg's Uncertainty Principle

Sessions: Dec 2024, June 2024, Dec 2023, June 2023, Nov 2022, June 2022 | **Frequency: 6 times**

Question Pattern: State and prove the uncertainty principle in detail; explain its structural/physical implications.

2. Schrödinger Wave Equation Derivations (Time-Dependent & Independent)

Sessions: Dec 2025, June 2025, June 2024, June 2023, June 2022 | **Frequency: 5 times**

Question Pattern: Formulate the spatial and temporal variations of matter waves.

3. Particle in a 1D Potential Box (Eigenvalues & Eigenfunctions)

Sessions: Dec 2025, Dec 2024, June 2024, Dec 2023, June 2022 | **Frequency: 5 times**

Question Pattern: Solve the boundary conditions to prove energy quantization and define wave functions.

Unit 2: Wave Optics

1. Newton's Rings Formation & Wavelength Determination

Sessions: Dec 2025, Dec 2024, June 2024, Dec 2023, June 2023, June 2022, Nov 2022 | **Frequency: 7 times**

Question Pattern: Explain ring construction; find expression for fringe diameters or liquid refractive indices. Includes standard numericals.

2. Michelson Interferometer Working & Fringe Shifts

Sessions: Dec 2025, June 2025, June 2024, Dec 2023, June 2022 | **Frequency: 5 times**

Question Pattern: Describe operational mechanisms with full setups and mirror displacement equations.

3. Mach-Zehnder Interferometer Architecture

Sessions: June 2025, June 2023, Nov 2022, June 2022 | **Frequency: 4 times**

Question Pattern: Outline construction and path splitting properties using proper ray layouts.

Unit 3: Semiconductor Physics

1. Hall Effect Experiment, Voltage, and Coefficients

Sessions: Dec 2025, June 2025, Dec 2024, June 2023, June 2022, Nov 2022 | **Frequency: 6 times**

Question Pattern: Derive Hall voltage, Hall angle, and present modern engineering applications.

2. Zener Diode Operational V-I Characteristics

Sessions: June 2025, June 2024, June 2023, June 2022 | **Frequency: 4 times**

Question Pattern: Graph the characteristic curve; explore Avalanche vs Zener breakdown boundaries.

3. Fermi Level Shifts in Extrinsic Semiconductors

Sessions: Dec 2025, Dec 2024, June 2024, Dec 2023, June 2022 | **Frequency: 5 times**

Question Pattern: Trace variation trends across doping profiles and temperature levels.

Unit 4: Lasers & Fiber Optics

1. Helium-Neon (He-Ne) Laser Setup & Transitions

Sessions: Dec 2025, Dec 2024, June 2024, Dec 2023, Nov 2022 | **Frequency: 5 times**

Question Pattern: Draw the schematic and explain gaseous atomic energy state processes.

2. Optical Fiber Parameter Derivations (Numerical Aperture & Acceptance Angle)

Sessions: Dec 2025, June 2025, Dec 2024, June 2024, Dec 2023, June 2022 | **Frequency: 6 times**

Question Pattern: Formulate acceptance cones from refractive indexes (n_1, n_2). Heavy core numerical target.

3. Einstein Coefficients Relationship

Sessions: Dec 2025, June 2025, Nov 2022 | **Frequency: 3 times**

Question Pattern: Balance radiative transitions to link A and B probability coefficients.

Unit 5: Electrodynamics & Vector Calculus

1. Current Density Continuity Equation

Sessions: June 2025, Dec 2024, June 2024, June 2023, June 2022 | **Frequency: 5 times**

Question Pattern: Model the time-rate of charge loss against current divergence.

2. Maxwell's Field Equations in Vacuum

Sessions: Dec 2025, June 2025, June 2024, Nov 2022 | **Frequency: 4 times**

Question Pattern: Frame the core electrodynamic equations assuming zero charge and current density space.

3. Divergence and Stokes Integral Vector Theorems

Sessions: Dec 2024, Dec 2023, June 2023, Nov 2022 | **Frequency: 4 times**

Question Pattern: Define and prove integration space equivalencies.